



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
13-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

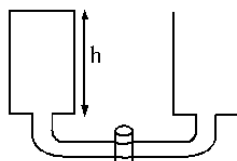
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Two identical cylindrical tanks are connected by a narrow tube with a cork at its middle (fig). The radius of a tank is $R = 20.0$ cm, the radius of the tube is $r = 1.0$ mm. The length of the tube is $\ell = 1$ m. Water ($\rho = 1000$ kg/m³) is poured into one of the tanks to a height of $h = 50$ cm, while the second tank is empty. At the instant $t = 0$, the cork is opened. [Both the tanks are open at top]



(viscosity of water $\eta = 1 \times 10^{-3}$ Pa-s)

Given poiseuille eqn for volume flow rate of fluid in a tube

$$\frac{dV}{dt} = \frac{\Delta P \pi r^4}{8\eta L}$$

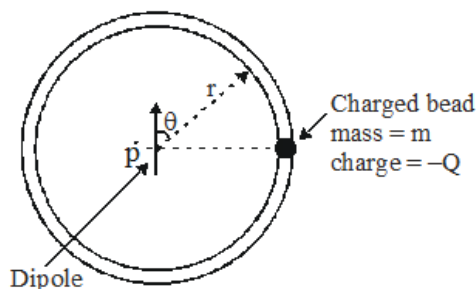
$\Delta P \rightarrow$ pressure difference across ends of the tube

$r \rightarrow$ radius of tube

$L \rightarrow$ length of tube

- (A) The difference between the levels of the water in the tanks decreases e times in time 1.6×10^4 s
- (B) The difference between the levels of the water in the tanks decreases e times in time 3.2×10^4 s
- (C) Total mechanical energy of the fluid is conserved during the process
- (D) Water level in the two containers cannot become equal in a finite time

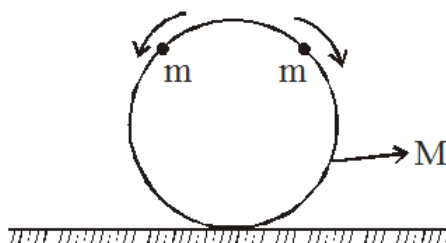
2. A small charged bead can slide on a circular frictionless, insulating wire frame. A point like dipole is fixed at the centre of circle, dipole moment is $\vec{p}$. Initially the bead is on the plane of symmetry of the dipole. Bead is released from rest. Ignore the effect of gravity. Mark the correct options



- (A) Magnitude of velocity of bead as function of its angular position is $\sqrt{\frac{Qp \cos \theta}{2\pi\epsilon_0 m r^2}}$
- (B) Normal force exerted by the string on bead is zero at all points
- (C) If the wire frame were not present bead executes circular motion and returns to initial point after tracing a complete circle.
- (D) Bead would move along a circular path until it reached the opposite its starting position and then executes periodic motion
3. A point charge $q = 6\mu\text{C}$ is moving in a straight line with a velocity $\vec{v} = 5 \times 10^4 \vec{i} \text{ (m/s)}$. When the charge is at the location P (3m, 4m, 0) choose the correct statements about the electric & magnetic fields produced by the charge at the origin.
- (A) Magnitude of magnetic field is $9.6 \times 10^{-10} \text{ T}$
- (B) Magnetic field is in $-z$ direction
- (C) Electric field is varying with time
- (D) Magnetic field is decreasing (with time) in magnitude

Space for Rough Work

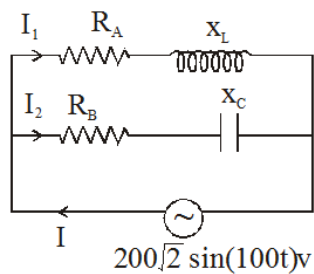
4. Two cars X and Y are moving with speed 15 m/s and 11 m/s respectively in opposite directions approaching each other from far. The driver in car X blows a horn which has components of frequencies ranging from 650 Hz to 800 Hz. The band width of frequencies is thus 150 Hz, speed of sound is 340 m/s. The correct statement(s) for observer in car Y is :-
- (A) The bandwidth of frequencies is 150 Hz
- (B) The bandwidth of frequencies is 162 Hz
- (C) Speed of sound of horn is 351 m/s
- (D) Speed of sound of horn is 329 m/s
5. Two beads of mass m are positioned at the top of a frictionless hoop of mass M and radius R , which stands vertically on ground. The beads are now given tiny kicks, and they slide down the hoop, one to the right and one to the left, as shown in figure. Then choose the correct options



- (A) The contact force from the ground immediately after the tiny kicks on the hoop is $(M + 2m)g$
- (B) The contact force from the ground on the hoop is $(M + 2m)g$ at the instant when radius vector makes an angle of $\cos^{-1}(2/3)$ with the upward vertical.
- (C) The smallest value of $\frac{m}{M}$ for which the hoop will rise off the ground at some time during the motion is $\frac{3}{2}$.
- (D) The hoop will never rise off the ground irrespective of the values of m and M .

Space for Rough Work

6. Figure shows a two branched parallel circuit with $R_A = 10\Omega$, $L = \frac{\sqrt{3}}{10}H$, $R_B = 20\Omega$ and $C = \frac{\sqrt{3}}{2}mH$. Current in $L - R_A$ is I_1 and in $C - R_B$ is I_2 and main current is I



- (A) Phase difference between I_1 and I_2 is 90°
- (B) At some instant current in $L-R_A$ is $10A$. At the same instant current in $C-R_B$ branch will be $5\sqrt{3}A$
- (C) At some instant I_1 is $10\sqrt{2}A$ then at this instant I will be $10\sqrt{2}A$
- (D) Power dissipated in the circuit is $2121.3W$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

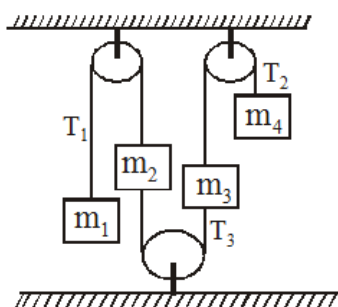
- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. The arrangement is kept in a vertical plane and all masses are released from rest with strings taut. In list-I different values of masses are given. Match with corresponding parameters in list-II. Symbols have their usual meanings. Pulleys and strings are ideal.



List-I					List-II	
	m_1	m_2	m_3	m_4		
(P)	2m	m	3m	4m	(1)	$ a_1 = a_3 $
(Q)	m	2m	4m	3m	(2)	$T_1 = T_2$
(R)	m	2m	3m	4m	(3)	$T_2 > T_1$
(S)	4m	3m	2m	m	(4)	$ a_1 > a_3 $
					(5)	$T_3 = 0$

(A) $P \rightarrow 1,3; Q \rightarrow 3,4; R \rightarrow 4,5; S \rightarrow 1,5$

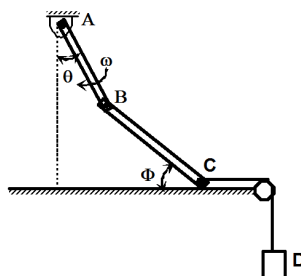
(B) $P \rightarrow 1,3; Q \rightarrow 3,4,5; R \rightarrow 3,4,5; S \rightarrow 5$

(C) $P \rightarrow 3,4; Q \rightarrow 3,4; R \rightarrow 1,5; S \rightarrow 2,3,4$

(D) $P \rightarrow 3,4; Q \rightarrow 2,4,5; R \rightarrow 1,3; S \rightarrow 4$

Space for Rough Work

8. Two light rods of length 1 m each are hinged together as shown in figure. Rod AB makes an angle θ with vertical while rod BC makes an angle ϕ with horizontal. End C of rod BC remains in contact with horizontal. Rod AB is rotated with constant angular velocity $\omega = 1$ rad/s in clockwise direction. At the instant when $\theta = 30^\circ$ and $\phi = 30^\circ$ match the variables in list-I with values in list-II.



List-I		List-II	
(P)	Angular velocity of rod BC in rad/s	(1)	$\frac{(3\sqrt{3} + 1)}{3\sqrt{3}}$
(Q)	Velocity of block D in m/s	(2)	$\frac{(\sqrt{3} - 1)}{\sqrt{6}}$
(R)	Magnitude of angular acceleration of rod BC in rad/s^2	(3)	$\frac{1}{\sqrt{3}}$
(S)	Acceleration of point B in m/s^2	(4)	1

(A) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

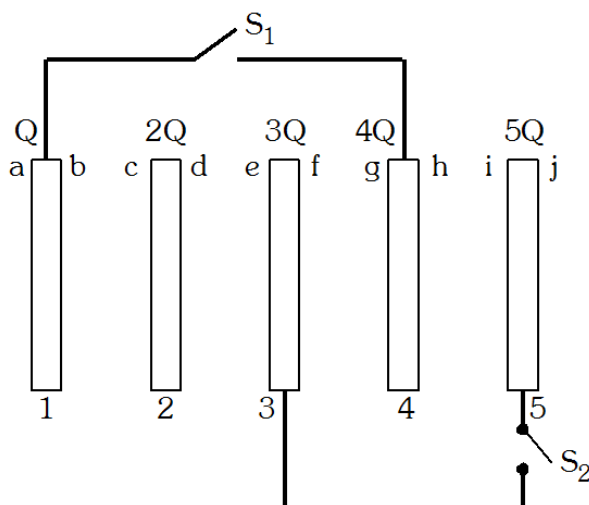
(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

Space for Rough Work

9. Consider a system of five large conducting plates of area A . The charge on plate 1, 2, 3, 4, 5 are given as Q , $2Q$, $3Q$, $4Q$ and $5Q$ respectively. Initially the two switches are opened. Area of each plate is A . The distance between every successive plate is very very small equal to d .



List-I		List-II	
(P)	The charge on surface b is	(1)	$-\frac{9}{2} \left(\frac{Qd}{A\epsilon_0} \right)$
(Q)	The potential difference between plate 2 and 3 ($V_2 - V_3$) is	(2)	$-0.6 Q$
Now both the switches S_1 and S_2 are closed simultaneously. Now Consider a new steady state.			
(R)	The charge on surface b is	(3)	$-1.3 Q$
(S)	The charge on surface f is	(4)	None of these

(A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

Space for Rough Work

10. Phase space diagrams are useful tools in analyzing all kinds of dynamical problems. They are especially useful in studying the changes in motion as initial position and momentum are changed. Here we consider some simple dynamical systems for which phase space is a plane in which position is plotted along horizontal axis and momentum is plotted along vertical axis. The phase space diagram is $x(t)$ vs. $p(t)$ curve in this plane. The arrow on the curve indicates the time flow. For example, the phase space diagram for a particle moving with constant velocity is a straight line as shown in the figure. Similarly we may also plot momentum of a pendulum versus θ (with sign convention shown in figure (b)).

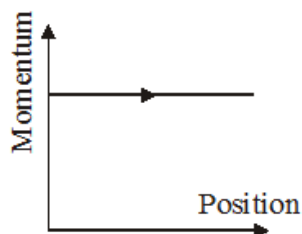


Figure (a)

Figure (b) shows phase diagram of motion of simple pendulum (momentum P versus angle θ). Choose potential energy level at the lowest point of the pendulum. E represents total energy of simple pendulum. Pendulum has a point mass connected with light rod.

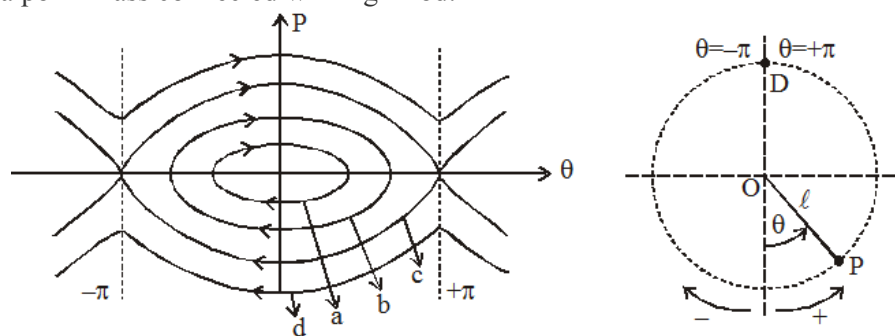


Figure (b)

List-I		List-II	
(P)	Phase diagram a	(1)	$E < 2mg\ell$
(Q)	Phase diagram b	(2)	$E \geq 2mg\ell$
(R)	Phase diagram c	(3)	May perform periodic and oscillatory motion
(S)	Phase diagram d	(4)	May represent SHM
		(5)	May represent angular velocity ω versus θ for a pendulum bob

- (A) $P \rightarrow 1,3,4,5; Q \rightarrow 1,3,4,5; R \rightarrow 2,5; S \rightarrow 2,5$ (B) $P \rightarrow 1,3,4,5; Q \rightarrow 1,4,5; R \rightarrow 1,5; S \rightarrow 1,5$
 (C) $P \rightarrow 4,5; Q \rightarrow 1,3,4,5; R \rightarrow 1,4,5; S \rightarrow 2,5$ (D) $P \rightarrow 2,4,5; Q \rightarrow 3,4,5; R \rightarrow 1,4; S \rightarrow 1,3,5$

Space for Rough Work

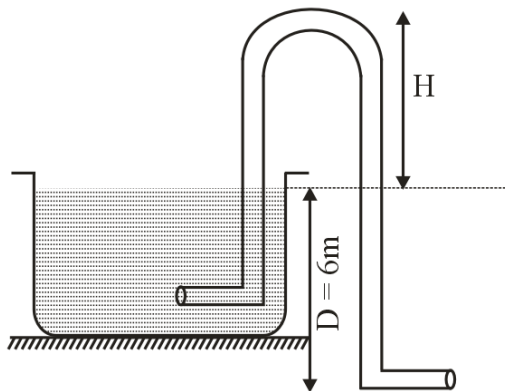
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

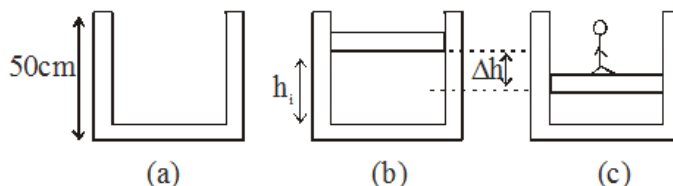
Zero Marks : 0 In all other cases.

1. In YDSE experiment two thin transparent sheets are used in front of the slits S_1 and S_2 , one of the thin sheet has refractive index $\mu_1 = 1.6$ and other has $\mu_2 = 1.4$. If both sheets have average thickness $\left(\frac{t_1 + t_2}{2}\right)$ the central maxima is at 5 mm from center O. Now the sheets are replaced by two sheets of same material of refractive index $\left(\frac{\mu_1 + \mu_2}{2}\right)$ but having thickness t_1 and t_2 now central maxima is observed at a distance of 8mm from center O. For $t_1 > t_2$ find the ratio $\frac{t_1}{t_2}$. Given $d = 1\text{mm}$, $D = 1\text{m}$
2. Figure shows a siphon, the vessel area is very large as compared to cross section of tube. Tube has a uniform cross section, its lower end is 6m below the surface of water. What is maximum height H (in m) of the upper end for siphon to work ? Take density of water = 10^3 kg/m^3 , atmospheric pressure = 10^5 N/m^2 .

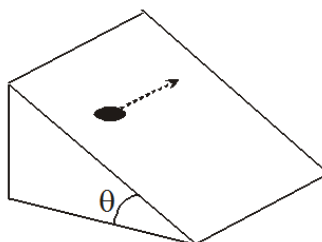


Space for Rough Work

3. A highly conducting cylinder that has cross-sectional area of 100 cm^2 and 50 cm deep is filled with air at 21°C and 1.00 atm figure (a). A 20 kg piston is now lowered into the cylinder, compressing the air trapped inside figure (b). Finally a 80 kg man stands on the piston, further compressing the air, which remains at 21°C figure (c). What temperature T (in $^\circ\text{C}$) should the gas be maintained to raise the piston and the man back to h_i . Take $g = 10 \text{ m/s}^2$



4. An object of mass $m = \sqrt{8} \text{ kg}$ rests on an inclined plane that makes angle $\theta = 45^\circ$ with the horizontal floor. What minimum force (in N), parallel to the base of the incline must be applied to the object in order to begin to move it along the plane parallel to the floor as shown? The coefficient of static friction between the object and the plane is $\mu_s = 1.25$.



5. A 3.5 V mobile phone battery can produce 1 A of current for 1 hour . This can be charged using a square solar panel 25 cm on each side. Assuming an efficiency of 10% and an incident solar power of 1 kWm^{-2} what time (in minutes) is needed to charge the battery?

Space for Rough Work

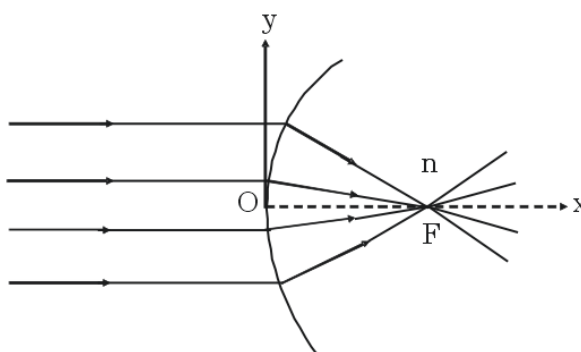
6. A parallel plate condenser, with plate area A and distance between plates d , is filled with a medium whose permittivity varies as ;

$$\epsilon(x) = \epsilon_0 + \beta\epsilon_0 x/d \quad 0 < x < \frac{d}{2}$$

$$\epsilon(x) = \epsilon_0 + \beta\epsilon_0 \frac{(d-x)}{d} \quad \frac{d}{2} < x < d.$$

x is the distance from one of the plates. For what value of β would the capacity of the condenser be $\frac{1}{2\ln 2}$ times that when it is completely filled with a uniform dielectric having dielectric constant β ?

7. Section of a refracting surface is shown in figure. The section is symmetrical about x axis. Refractive index of medium on right of the surface is n with respect to the medium on left of the surface. Parallel monochromatic light rays are incident on the surface as shown in figure and the refracted rays are focused at F at a distance f from origin O of the co-ordinate system shown in figure. The equation $y(x)$ of shape of the section shown is found to be $(n^2 - 1)x^2 + n^2y^2 - Bn(n-1)fx = 0$. Find B .



8. In the determination of Young's modulus by using Searle's method, a wire of length $L = 2\text{m}$ and diameter $d = 0.5\text{ mm}$ is used. For a load $M = 2.5\text{ kg}$, an extension $\ell = 0.25\text{ mm}$ in the length of the wire is observed. Quantities d and ℓ are measured using a screw gauge and a micrometer, respectively. They have the same pitch of 0.5 mm . The number of divisions on their circular scale is 100. The length of wire, acceleration due to gravity and mass M are known exactly. The percentage error in measurement of the Young's modulus is _____

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Correct statement :

- (A) Starch is the plant storage of carbohydrate.
(B) Natural rubber is the trans-polymer.
(C) Nylon show positive Lassigne test with Na metal.
(D) Bakelite is phenol-acetaldehyde resin.

2. Consider the following values of I.E.(eV) for elements W and X:

Element	I.E. ₁	I.E. ₂	I.E. ₃	I.E. ₄
W	10.5	15.5	24.9	79.8
X	8	14.8	78.9	105.8

Other two element Y and Z have outer electronic configuration ns^2np^4 and ns^2np^5 respectively. Then according to given information which of the following compound(s) is/are not possible.

- (A) W_2Y_3 (B) X_2Y_3 (C) WZ_2 (D) XZ_2

Space for Rough Work

3. (I) $\text{Na} + \underset{\text{excess}}{\text{O}_2} \xrightarrow{\Delta}$ major product
 (II) $\text{K} + \underset{\text{excess}}{\text{O}_2} \xrightarrow{\Delta} ?$
 (III) $\text{O}_2 + \text{PtF}_6 \xrightarrow{\Delta}$ red compound prepared by Neil Bartlett.

select correct for above reactions

- (A) Electron population in p^* molecular orbitals of O_2 increases in (I) and (II)
 (B) Electron population in p^* molecular orbitals of O_2 decreases in (III)
 (C) π bond order of O–O bond is lowest in product of IIIrd reaction
 (D) Fractional O–O bond order is observed in product of Ist reaction.
4. Which of the following polymer is/are Biodegradable?

- (A) $\text{-(CH}_2\text{-}\underset{\text{Cl}}{\text{C}}\text{=CH-CH}_2\text{)-}_n$
 (B) $\text{-(CH}_2\text{-CH=CH-CH}_2\text{-CH}_2\text{-}\overset{\text{CN}}{\text{CH}}\text{)-}_n$
 (C) $\text{-(O-}\underset{\text{CH}_3}{\text{CH}}\text{-CH}_2\text{-}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{-O-}\underset{\text{CH}_2\text{CH}_3}{\text{CH}}\text{-CH}_2\text{-}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{)-}_n$
 (D) $\text{-(}\overset{\text{H}}{\text{N}}\text{-(CH}_2\text{)}_6\text{N-}\overset{\text{H}}{\text{C}}\text{(=O)-}\underset{\text{O}}{\text{C}}\text{(CH}_2\text{)}_4\text{C(=O)-)}_n$

5. Select correct statement(s)
 (A) Plaster of Paris is a hemihydrate of CaSO_4 obtained by heating the gypsum around 300 K.
 (B) Sodium carbonate is used in hard-water softening.
 (C) Sodium peroxide reacts with cold water giving hydrogen peroxide
 (D) The mobilities of the alkali metal ions in aqueous solutions are $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$
6. In a hypothetical solid, "C" atoms forming cubical closed packed lattice. "A" atoms occupy all tetrahedral void and "B" atoms occupy all octahedral voids. There is no distortion in ccp lattice. Fraction of body diagonal not covered up by atoms is :
 (A) 0.76 (B) 0.24 (C) 0.68 (D) 0.12

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

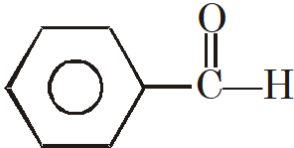
- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.

	List-I		List-II
(P)	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$	(1)	Disproportionation reaction with OH^-
(Q)	$\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$	(2)	Condensation reaction with OH^-
(R)	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$	(3)	Silver mirror test
(S)		(4)	Haloform test

(A) $\text{P} \rightarrow 1,3; \text{Q} \rightarrow 2,4; \text{R} \rightarrow 1,3; \text{S} \rightarrow 2,3$

(B) $\text{P} \rightarrow 2,3,4; \text{Q} \rightarrow 1,3; \text{R} \rightarrow 2,4; \text{S} \rightarrow 1,3$

(C) $\text{P} \rightarrow 3,4; \text{Q} \rightarrow 2,4; \text{R} \rightarrow 1,2,3; \text{S} \rightarrow 1,2,3$

(D) $\text{P} \rightarrow 1,2,3; \text{Q} \rightarrow 2,3; \text{R} \rightarrow 3,4; \text{S} \rightarrow 2,3$

Space for Rough Work

8.

	List-I		List-II
(P)	Fe^{n+}	(1)	(For $n = 0$), forms white ppt with dil. HCl.
(Q)	Cu^{n+}	(2)	(For $n = 2$), forms coloured ppt with excess of NaOH solution.
(R)	Hg^{n+}	(3)	(For $n = 0$), N_2O gas will be released with dil. HNO_3 .
(S)	Pb^{n+}	(4)	(For $n = 2$), Forms soluble complex with excess of NH_3 solution.
		(5)	(For $n = 2$) forms black ppt on passing H_2S gas in its aqueous

 (A) $\text{P} \rightarrow 2,3; \text{Q} \rightarrow 2,4,5; \text{R} \rightarrow 2,5; \text{S} \rightarrow 1,5$

 (B) $\text{P} \rightarrow 1,2,3; \text{Q} \rightarrow 3,5; \text{R} \rightarrow 1,2,5; \text{S} \rightarrow 2,4,5$

 (C) $\text{P} \rightarrow 1,4; \text{Q} \rightarrow 2,4; \text{R} \rightarrow 3,5; \text{S} \rightarrow 1,3$

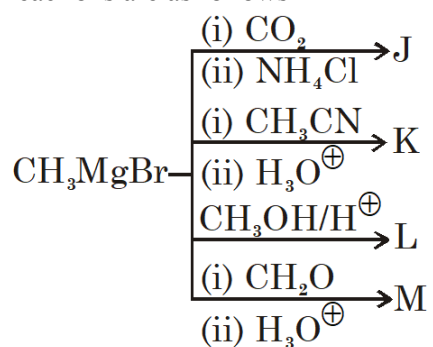
 (D) $\text{P} \rightarrow 2,4,5; \text{Q} \rightarrow 2,3,5; \text{R} \rightarrow 1,5; \text{S} \rightarrow 1,4,5$

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9. Complete the following reactions identify the major products and apply a chemical test to distinguish between following pairs :

	List-I (Pairs)		List-II (Chemical Test)
(P)	J and M	(1)	Na-metal
(Q)	J and K	(2)	NaOI
(R)	K and M	(3)	2,4-DNP
(S)	L and M	(4)	NaHSO ₃

Reactions are as follows



(A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

Space for Rough Work

10.

	List-I		List-II
(P)	<chem>Nc1ccccc1.CCl(Cl)Cl>>[KOH][\Delta]</chem>	(1)	Carbanion is intermediate
(Q)	<chem>[D]C([D])([D])C(=O)Nc1ccccc1>>[KOBr]</chem>	(2)	Carbene is intermediate
(R)	<chem>Oc1ccccc1>>[(i)CCl(Cl)Cl/NaOH][(ii)[H+]]</chem>	(3)	Product containing N-atom
(S)	<chem>C=C([N+])([O-])>>[H2SO4][\Delta]</chem>	(4)	Cyanide or isocyanide is product

(A) P → 1,3,4; Q → 3,4; R → 2,3; S → 2,3,4

(B) P → 1,2,3,4; Q → 3; R → 1,2; S → 3,4

(C) P → 2; Q → 1,3,4; R → 2,3; S → 1,2

(D) P → 3,4; Q → 3,4; R → 2,3; S → 2,3

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Find total number of species in which incoming electron enters into H.O.M.O.
 $H_2, Li_2, B_2, C_2, N_2, O_2, F_2, Na_2, S_2, Cl_2$.
2. 40 ml of 0.05 M solution of sodium sesquicarbonate ($Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$) is titrated against 0.05 M HCl. "X" ml of HCl is used when phenolphthalein is the indicator and "Y" ml of HCl is used when methyl orange is the indicator in two separate titrations. Hence, value of $\frac{(Y - X)}{10}$ is
3. Following reaction takes place at $\left(\frac{600}{2.303 \times 0.8314} \right)$ K temperature.
 $H_2(g) + 2Ag^+(aq) \rightleftharpoons 2Ag(s) + 2H^+(aq)$
 $P_{H_2} = 1.0 \text{ bar}, [Ag^+] = 10^{-6} \text{ M}, [H^+] = 10^{-3} \text{ M}, \Delta_f G^\circ(Ag^+, aq) = 75 \text{ kJ mol}^{-1}$
 Calculate z.
 where $z = \frac{|\Delta_r G(\text{in kJ})|}{19}$
4. Select the number of ores for which roasting process is applied during metallurgy.
 Argentite, Copper pyrites, Galena, Zinc blende, Chalcocite

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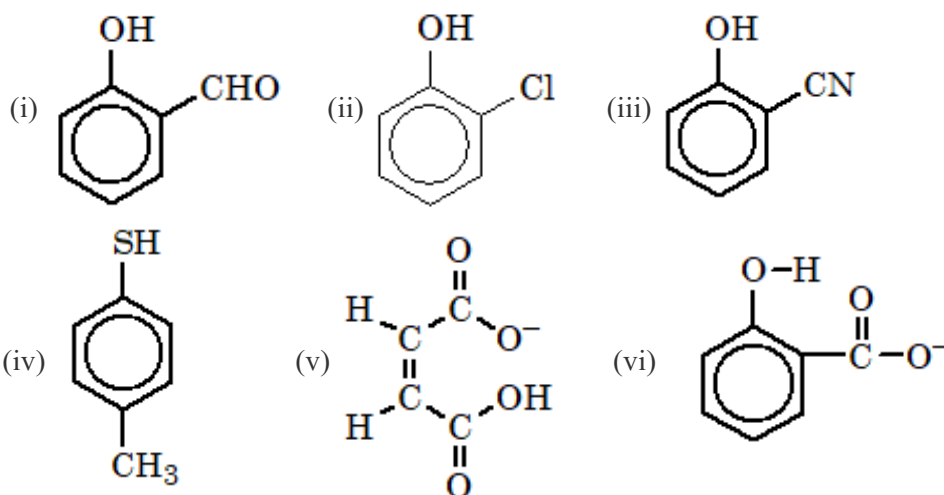
5. In the saturated aqueous solution of PbCl_2 the freezing point decreases by $\left(\frac{X}{100}\right)^\circ\text{C}$ then 'X' is :

(Given k_{sp} of $\text{PbCl}_2 = 4 \times 10^{-6}$, $K_{f, \text{water}} = 2 \text{ k kg/mole}$)

6. Total number of compounds which are soluble in a hot aqueous NaOH are:

- (i) Salicyclic acid
- (ii) Aspirine
- (iii) Formic acid
- (iv) Acetic acid
- (v) Succinic anhydride
- (vi) Cyclohexanone
- (vii) Benzene sulphonamide
- (viii) Cyclohexene

7. In how many compounds intermolecular H-bonding will exist.



8. A sample of H-atoms containing all the atoms in a particular excited state, absorb radiations of a particular wave length by which the atoms get excited to another excited state. When the atoms finally de-excite to the ground state, they emit radiations of 15 different wavelength. Out of these 15 radiations, 5 have wavelengths shorter than the absorbed radiation and 9 have wavelength longer than the absorbed radiation. What is the initial excited state of atoms?

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $f(x) = 2x^3 - 15x^2 + 36x - 23$ and $g(x) = \begin{cases} \max\{f(t), 1 \leq t \leq x\}, & \text{if } 1 \leq x \leq \frac{7}{2} \\ 12 - 2x, & \text{if } \frac{7}{2} < x \leq 6 \end{cases}$. Which is correct about $g(x)$

(A) Continuous everywhere

(B) Differentiable everywhere

(C) has greatest value 5

(D) None of these

2. For distinct complex numbers $z_1, z_2, \dots, z_n$, ($n > 2$) the value of $\frac{|z_2 - z_1|^2 + |z_3 - z_2|^2 + \dots + |z_n - z_{n-1}|^2}{|z_n - z_1|^2}$ can not be less than

(A) $\frac{1}{n-1}$

(B) $\frac{1}{n}$

(C) $\frac{1}{n+1}$

(D) $\frac{1}{n+2}$

Space for Rough Work

3. Equation(s) of possible common tangents to $y^2 = 8(x - 3)$ and $x^2 = 8(y - 3)$ is/are :
- (A) $x + y = 1$ (B) $2x - y = 5$
 (C) $x - 2y + 5 = 0$ (D) None of these
4. If the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ intersects line $3\beta^2x + 3(1 - 2\alpha)y + z = 3 = -\frac{1}{2}(6\alpha^2x + 3(1 - 2\beta)y + 2z)$ then point $(\alpha, \beta, 1)$ lies on the plane
- (A) $2x - y + z = 4$ (B) $x + y - z = 2$
 (C) $x - 2y = 0$ (D) $2x - y = 0$
5. Let $\int_0^\infty e^{-x^2} dx = a$ and $I_n = \int_0^\infty x^n e^{-x^2} dx$, $n \in \mathbb{N}$, then
- (A) $I_n = \left(\frac{n-1}{2}\right) I_{n-2}$ (B) $I_n = \frac{(n-1)!}{2^{n-1} \left(\frac{n-2}{2}\right)!} a$, if n is even
 (C) $I_n = \frac{1}{2} \left(\frac{n-2}{2}\right)! a$, if n is even (D) $I_n = \frac{(n-1)!}{2^{n-1} \left(\frac{n-2}{2}\right)!}$, if n is even
6. In $\triangle ABC$ (with usual notation), if $\cos A + \cos B = 4\sin^2 \frac{C}{2}$, then which of the following hold(s) good ?
- (A) $\cot \frac{A}{2} \cot \frac{B}{2} = 2$ (B) $\cot \frac{A}{2} \cot \frac{B}{2} = 3$
 (C) a, c, b are in A.P. (D) a, b, c are in G.P.

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SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

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Negative Marks : -1 In all other cases

7. Match the following List-I with List-II

List-I		List-II	
(P)	If $A = [a_{ij}]_{3 \times 3}$ and $a_{ij} = i^2 + j^2$, then A is	(1)	singular
(Q)	$A = [a_{ij}]_{3 \times 3}$ and $a_{ij} = 3^{i-j}$, then A is	(2)	skew-symmetric
(R)	$A = [a_{ij}]_{3 \times 3}$ and $a_{ij} = i^2 - j^2$, then A is	(3)	symmetric
(S)	$A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$, then A is	(4)	idempotent

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 1; S \rightarrow 2$

(B) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 1$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 1$

Space for Rough Work

8. List-I with List-II (where $[.]$ denotes GIF and $\{.\}$ denotes fractional part function)

List-I		List-II	
(P)	For $f(x) = \begin{cases} \frac{[x-2]}{x-2} [\log_e x], & 1 \leq x \leq 3 - \{2\} \\ \{x^2\}, & 3 < x \leq \frac{7}{2} \end{cases}$ number of points of discontinuity is	(1)	0
(Q)	For $f(x) = \begin{cases} (\sqrt{2} + \sin \frac{1}{x}) e^{\frac{-1}{ x }} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ number of points of extremum is	(2)	5
(R)	number of points where $f(x) = \max \{ 3 - x^2 , 3 - x^3\}$ is not differentiable	(3)	2
(S)	$2 \lim_{x \rightarrow 0} x^2 \left(1 + 2 + 3 + \dots + \left[\frac{1}{ x } \right] \right) = ,$ where $[x]$ represents GIF	(4)	1

(A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

Space for Rough Work

9. Match the following :

List-I		List-II	
(P)	If $\sin^{-1} \cos x - \cos^{-1} \sin x = a$ has atleast one solution, then the exhaustive set in which 'a' can lie	(1)	$\{0\}$
(Q)	If equation $\sin^{-1}\sqrt{x} + \cos^{-1}\sqrt{x^2 - 1} + \tan^{-1}\tan y = a$ has atleast one solution, then the exhaustive set in which 'a' can lie	(2)	$\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$
(R)	If $\sqrt{ \sin^{-1} \sin x } + \sqrt{ \tan^{-1} \tan x } = 2\sqrt{a}$ has atleast one solution, then the exhaustive set in which 'a' can lie	(3)	$\left[0, \frac{\pi}{2}\right)$
(S)	$\sin^{-1}(x^2 + y^2) + \tan^{-1}\sqrt{4y^2 - 1} + \sec^{-1}x = a$, then the exhaustive set in which 'a' can lie	(4)	ϕ

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

Space for Rough Work

10. List-I with List-II

List-I		List-II	
(P)	Vertices of a triangle are (1, 2), (3, 4) & (2, -4)	(1)	$\lambda \neq -7$
(Q)	Equation of sides of a triangle are $x + y = 3$ $3x - 5y + 7 = 0$ and $x - 3y + 1 = 0$	(2)	triangle is obtuse angled
(R)	Orthocenter and circumcenter are on different half plane w.r.t larger side.	(3)	(1, 1) lies inside the triangle
(S)	Equations of sides of a triangle are, $3x - 4y - 13 = 0$, $8x - 11y - 33 = 0$ & $2x - 3y + \lambda = 0$	(4)	centroid lies on $3y = 2$

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(B) $P \rightarrow 2,4; Q \rightarrow 2,3; R \rightarrow 2; S \rightarrow 1,4$

(C) $P \rightarrow 2,4; Q \rightarrow 2,3,4; R \rightarrow 2; S \rightarrow 1,2$

(D) $P \rightarrow 2,4; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1,2,4$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

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1. The value of the expression ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$, for $n = 6$ is
2. A sequence is obtained by deleting all perfect squares from set of natural number. The remainder when the 2003rd term of new sequence is divided by 2048, is ?
3. Let $2(f(x))^2 - \frac{d^2f(x)}{dx^2} f(x) + \left(\frac{df(x)}{dx}\right)^2 = 0$ & $f(0) = f(1) = -1$. Area of region bounded by $y = 0$, $x = 0$, $x = 1$ and $y = (2x - 1) f(x)$ is $2 \left(\frac{e^{1/a} - 1}{e^{1/a}} \right)$, then 'a' is ?

Space for Rough Work

4. Two boys A and B find the jumble of n ropes lying on the floor. Each takes hold of one loose end randomly. If the probability that they are both holding the same rope is $\frac{1}{101}$ then the number of ropes is equal to ?
5. A circle of radius $r = 4$ units is inscribed in an equilateral triangle ABC, then an equilateral triangle is inscribed in the circle, a circle again is inscribed in the later triangle and so on. In this way the process continues infinitely. If $r, x_1, x_2, \dots, x_n, \dots$ be the radii of these circles respectively, then the sum of radii of all the circles $x_1 + x_2 + x_3 \dots \infty$ is equals to
6. A trapezium is inscribed in the parabola $y^2 = 4x$ such that its diagonals pass through the focus and are of length $\frac{25}{4}$ units. If area of this trapezium is A , then $4A$ is ?
-

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7. If α, β are the roots of the quadratic equation, $x^2 - 2p(x - 4) - 15 = 0$, then the greatest integer p for which one root is less than 1 and the other root is greater than 2 is ?
8. Let $\vec{A}, \vec{B}, \vec{C}$ be vectors of length 3, 4, 5 respectively. Let $\vec{A}$ be perpendicular to $\vec{B} + \vec{C}$, $\vec{B}$ is perpendicular to $\vec{C} + \vec{A}$ and $\vec{C}$ is perpendicular to $\vec{A} + \vec{B}$ and if the length of vector $|\vec{A} + \vec{B} + \vec{C}|$ is $k\sqrt{2}$ then find the value of k
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Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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